

Splunk SIEM – VPN Log Ingestion & Security Analysis

Tool: Splunk Enterprise

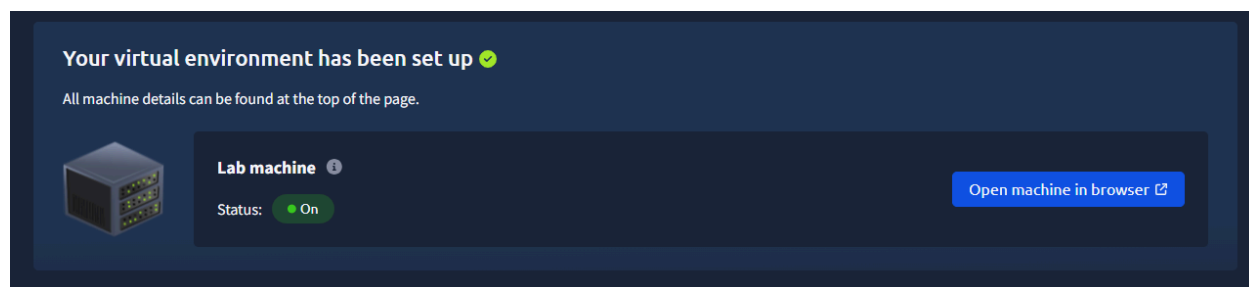
Environment: TryHackMe Lab

Project Focus: SIEM | Log Ingestion | SPL | Log Analysis | Security Investigation

1. Project Objective

The objective of this project was to understand how Splunk can be used in a SOC environment to collect, process, search, and investigate security logs.

I worked with **VPN logs in JSON format** and used Splunk to identify user activity, investigate source IP addresses, and analyze geographic information within the logs.



2. Understanding the Splunk Architecture

Before starting the investigation, I reviewed how the main Splunk components work together:

- **Forwarder** – collects logs from monitored systems and forwards them to Splunk.
- **Indexer** – processes, parses, and stores the incoming events.
- **Search Head** – allows analysts to search and investigate the indexed data using SPL.

This helped me understand the basic data flow from a log source to the analyst's search interface.

Log Source → Forwarder → Indexer → Search Head → Analyst Investigation

3. Preparing the VPN Log Data

The dataset provided for the investigation contained **VPN activity in newline-delimited JSON format**.

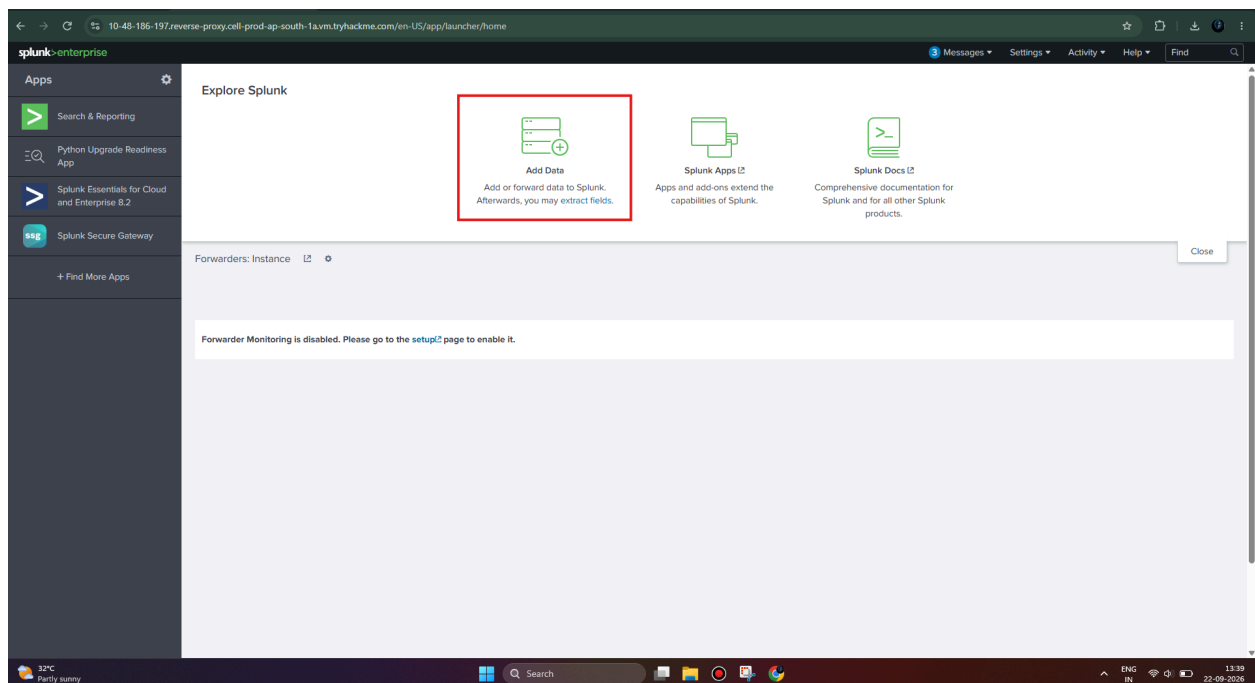
Before analyzing the data, I identified the important fields available in the logs, including:

- Username
- Source IP
- Source Country
- VPN activity information

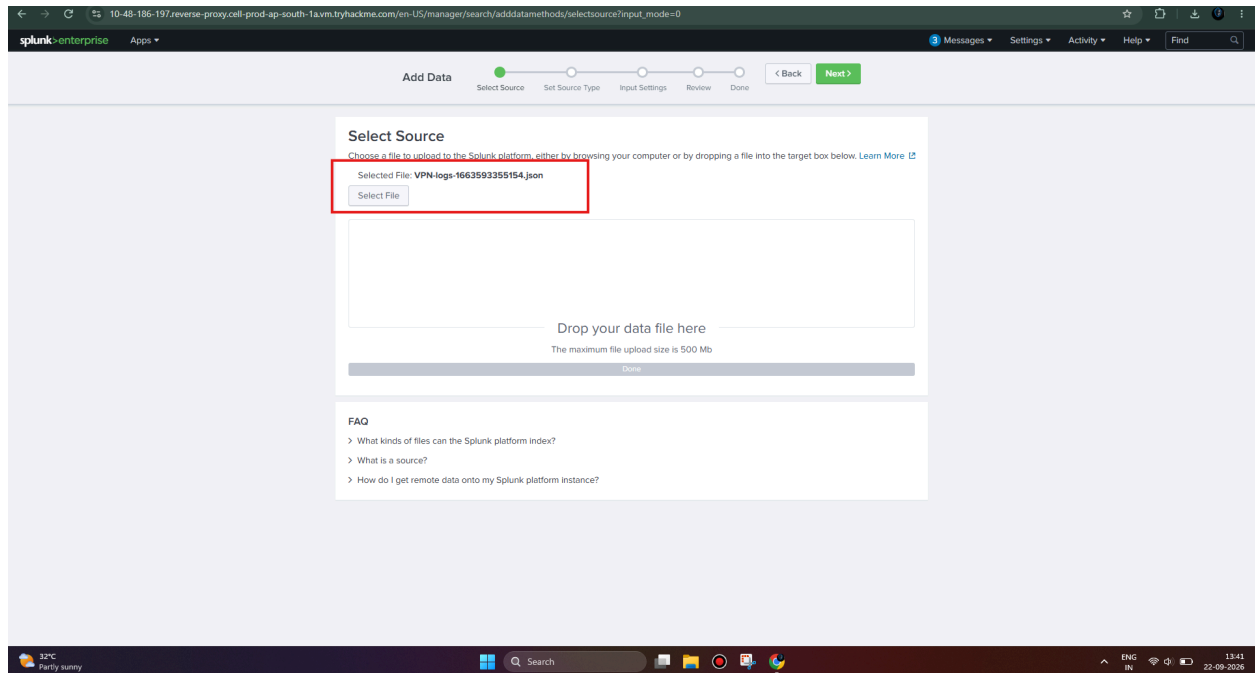
This provided the foundation for the investigation.

4. Ingesting the Logs into Splunk

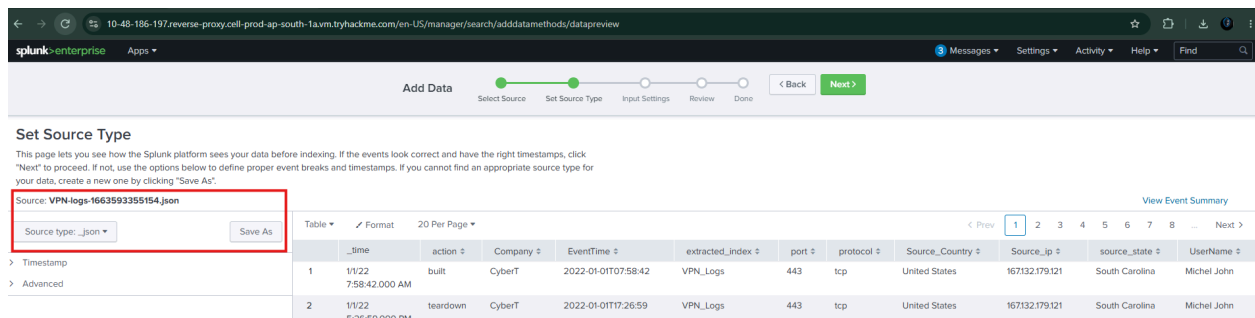
I used Splunk's **Add Data** → **Upload** workflow to ingest the VPN logs.



Selecting the VPN log file.



Confirming the appropriate **JSON** source type.



Configuring the input and host information.

← → 10-48-186-197.reverse-proxy.cell-prod-ap-south-1a.vmltyhackme.com/en-US/manager/search/adddatamethods/inputsettings

splunk>enterprise Apps

Messages Settings Activity Help Find

Add Data

Select Source Set Source Type Input Settings Review Done

< Back Review >

Input Settings

Optionally set additional input parameters for this data input as follows:

Host

When the Splunk platform indexes data, each event receives a "host" value. The host value should be the name of the machine from which the event originates. The type of input you choose determines the available configuration options. [Learn More](#)

☒ Constant value
☐ Regular expression on path
☐ Segment in path

Host field value: VPN_Connections

Index

The Splunk platform stores incoming data as events in the selected index. Consider using a "sandbox" index as a destination if you have problems determining a source type for your data. A sandbox index lets you troubleshoot your configuration without impacting production indexes. You can always change this setting later. [Learn More](#)

Index: Default Create a new index

Creating/selecting the VPN_Logs index.

New Index

General Settings

Index Name

VPN_Logs

Set index name (e.g., INDEX_NAME). Search using index=INDEX_NAME.

Index Data Type

Events

Metrics

The type of data to store (event-based or metrics).

Home Path

optional

Hot/warm db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/db).

Cold Path

optional

Cold db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/colddb).

Thawed Path

optional

Thawed/resurrected db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/thaweddb).

Data Integrity Check

Enable

Disable

Enable this if you want Splunk to compute hashes on every slice of your data for the purpose of data integrity.

Max Size of Entire Index

500

GB

Maximum target size of entire index.

Max Size of Hot/Warm/Cold Bucket

auto

GB

Maximum target size of buckets. Enter 'auto_high_volume' for high-volume indexes.

Frozen Path

optional

Frozen bucket archive path. Set this if you want Splunk to automatically archive frozen buckets.

Save

Cancel

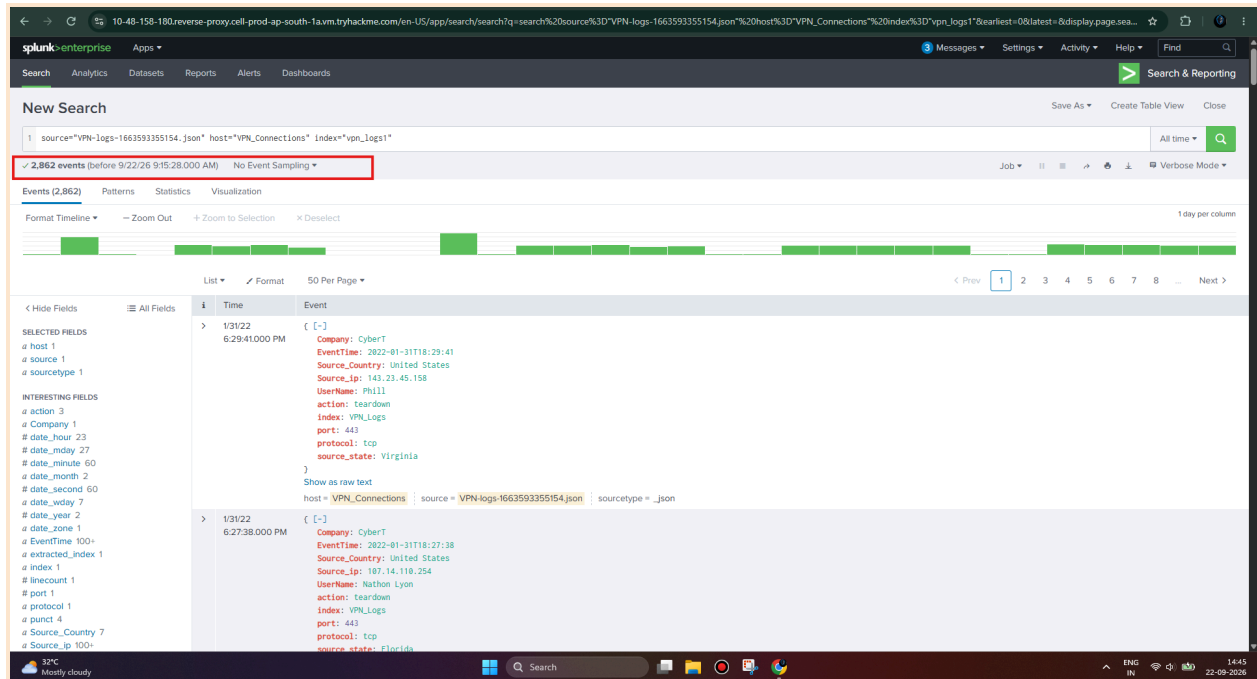
Reviewing the configuration and, after submitting, confirming the upload process is completed. The logs are available for searching in Splunk.

5. Validating Log Ingestion

Before beginning the investigation, I verified that the data had been successfully indexed.

I used:

```
source="VPN-logs-1663593355154.json" host="VPN_Connections" index="vpn_logs"
```



This allowed me to confirm that events were present in the **VPN_Logs** index.

I also set the Splunk time range to **All Time** to ensure the complete dataset was included during the analysis.

6. Extracting JSON Fields

Since the VPN logs were stored as JSON events, I used the **spath** command to extract the individual fields.

I used:

```
source="VPN-logs-1663593355154.json" host="VPN_Connections" index="vpn_logs1"
| spath
```

New Search

1 source='VPN-logs-1663593355154.json' host='VPN_Connections' index='vpn_logs1'
2 | spath

✓ 2,862 events (before 9/22/26 9:28:04.000 AM) No Event Sampling

Events (2,862) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect

1 day per column

List Format 50 Per Page

< Hide Fields All Fields

SELECTED FIELDS

- # host 1
- # source 1
- # sourcetype 1

INTERESTING FIELDS

- # action 3
- # Company 1
- # date_hour 23
- # date_mday 27
- # date_minute 60
- # date_month 2
- # date_second 60
- # date_wday 7
- # date_year 2
- # date_zone 1
- # EventTime 100+
- # extracted_index 1
- # index 2
- # linecount 1
- # port 1
- # protocol 1
- # punct 4
- # Source_Country 7

i	Time	Event
>	1/31/22 6:29:41.000 PM	<pre>{ "Company": "CyberT", "EventTime": "2022-01-31T18:29:41", "Source_Country": "United States", "Source_ip": "143.23.45.158", "Username": "Phill", "action": "teardown", "index": "VPN_Logs", "port": 443, "protocol": "tcp", "source_state": "Virginia" }</pre> Show as raw text host = VPN_Connections source = VPN-logs-1663593355154.json sourcetype = json
>	1/31/22 6:27:38.000 PM	<pre>{ "Company": "CyberT", "EventTime": "2022-01-31T18:27:38", "Source_Country": "United States", "Source_ip": "107.14.110.254", "Username": "Nathan Lyon", "action": "teardown", "index": "VPN_Logs", "port": 443, "protocol": "tcp" }</pre>

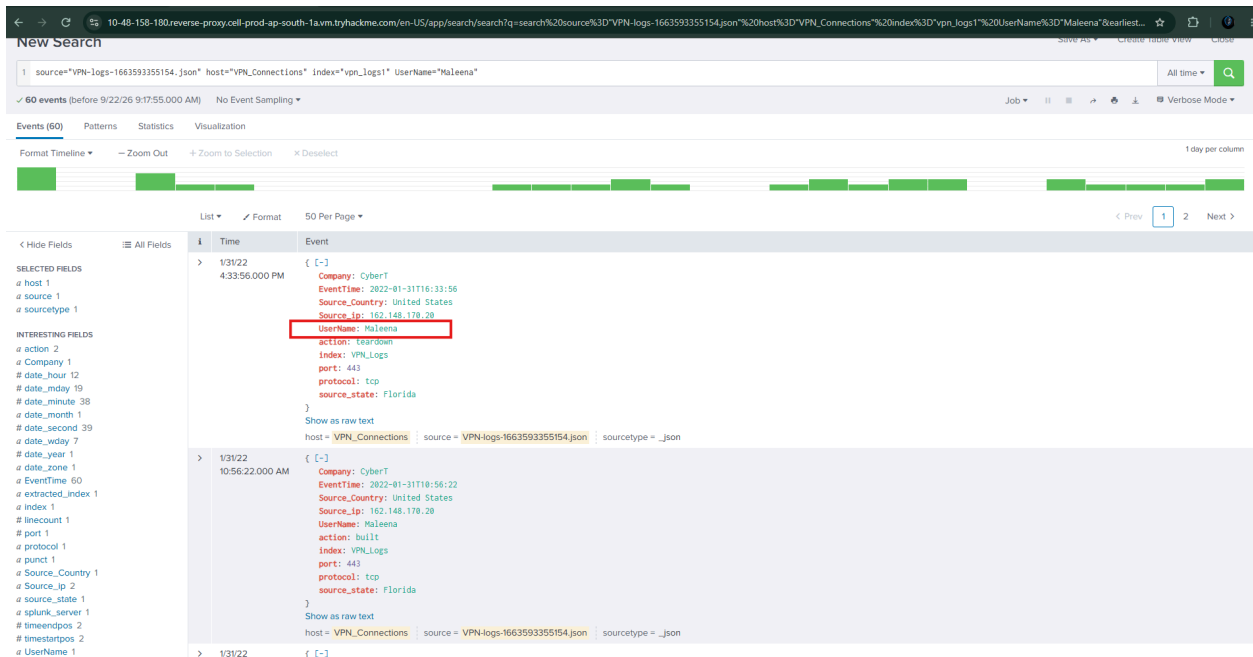
This allowed fields such as **UserName**, **Source_ip**, and **Source_Country** to be searched and analyzed more effectively.

7. Investigating User Activity

I then investigated activity associated with specific users.

I used:

```
source="VPN-logs-1663593355154.json" host="VPN_Connections" index="vpn_logs"
UserName="Maleena"
```



This query helped determine the number of VPN events associated with the selected user.

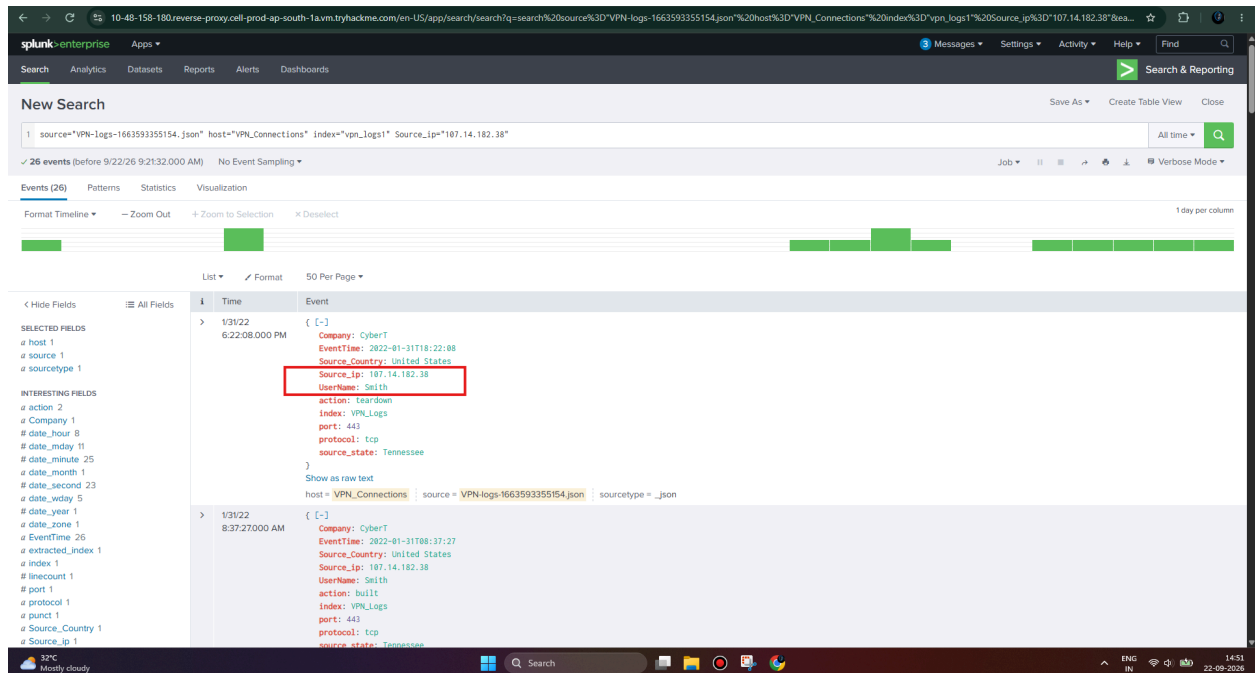
From a SOC perspective, this type of search can help an analyst understand **who generated the activity and how frequently it occurred**.

8. Investigating Source IP Activity

Next, I investigated activity associated with a specific source IP address.

I used:

```
source="VPN-logs-1663593355154.json" host="VPN_Connections" index="vpn_logs"
Source_ip="107.14.182.38"
```

This helped identify the users associated with the IP address and summarize the number of related events.

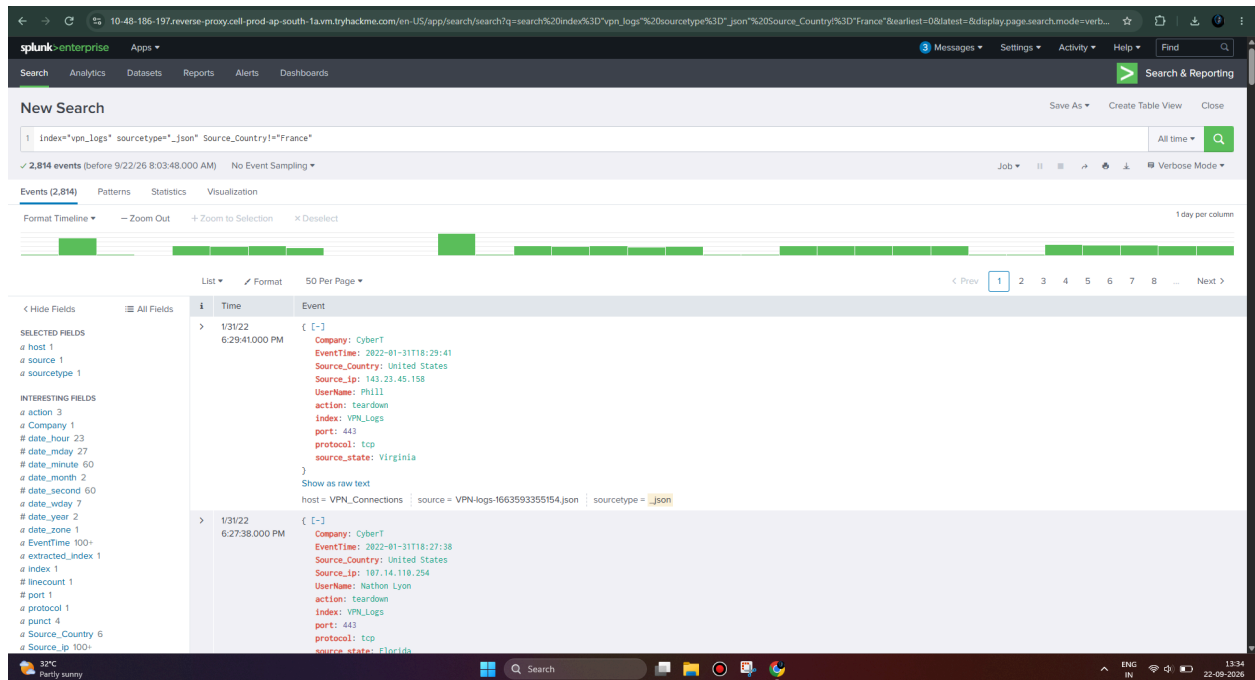
In a real SOC investigation, source-IP analysis can provide useful context when investigating suspicious authentication or VPN activity.

9. Reviewing Geographic Activity

I also analyzed the geographic information associated with VPN connections.

I used:

```
source="VPN-logs-1663593355154.json" host="VPN_Connections" index="vpn_logs"
Source_Country!="France"
```



This helped identify VPN events originating from countries outside the expected location.

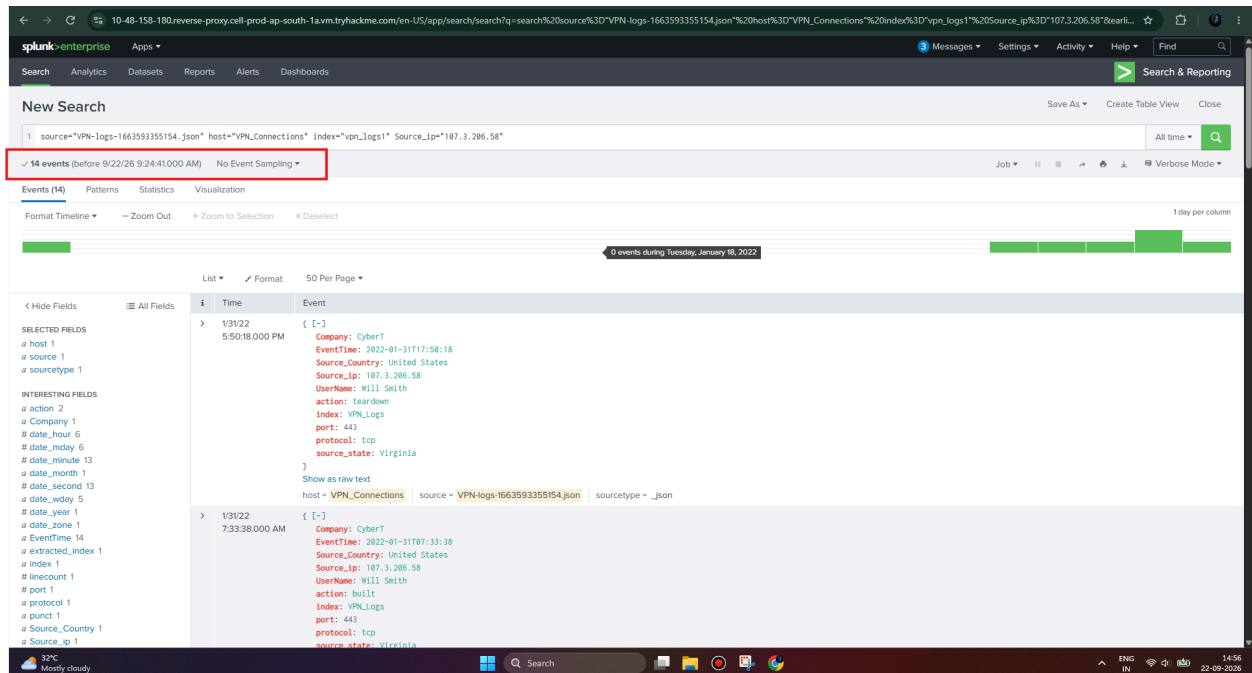
Geographic information can provide additional context during investigations, particularly when an account appears to be connecting from unexpected locations.

10. Investigating a Specific IP

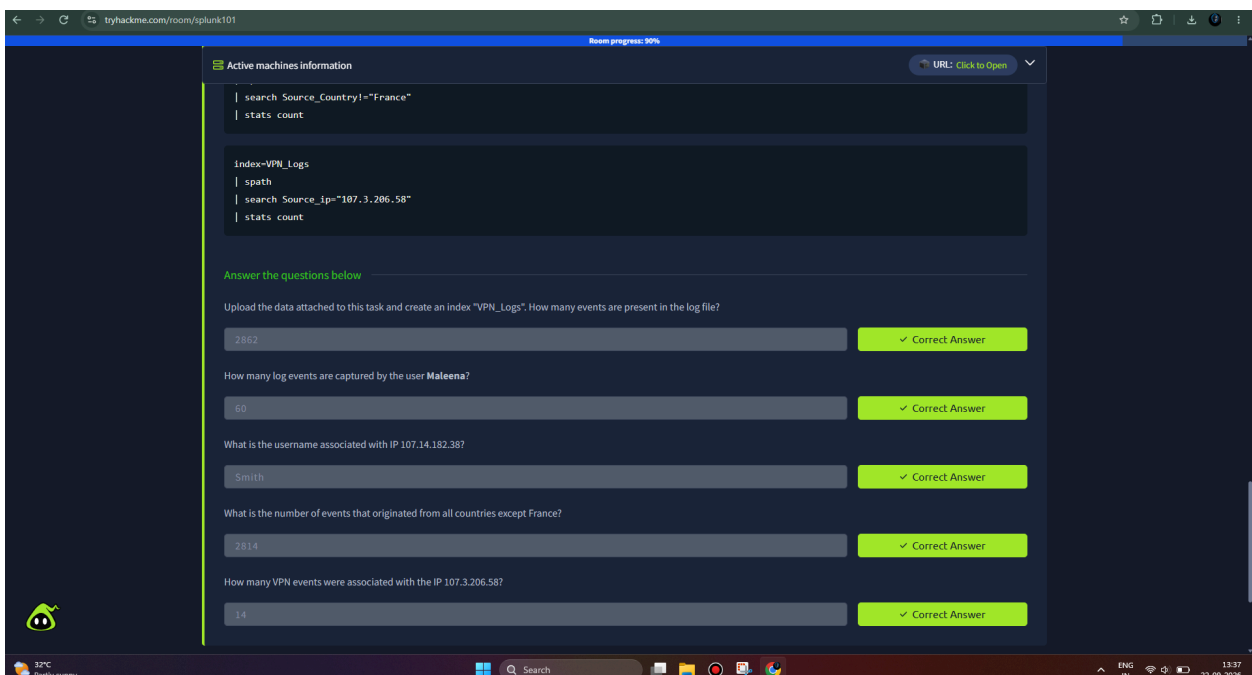
Finally, I investigated another source IP to determine how many VPN events are associated with the specific IP.

I used:

```
source="VPN-logs-1663593355154.json" host="VPN_Connections" index="vpn_logs"
Source_ip="107.3.206.58"
```



11. Investigation Workflow



The overall workflow I followed was:

Ingest Logs → Validate Data → Extract Fields → Search Events → Filter Activity → Investigate Users/IPs → Analyze Geographic Information → Correlate Findings

This helped me understand how a SOC analyst can move from raw log data toward useful investigation context.

12. Key Learning

This project gave me practical exposure to using **Splunk as a SIEM for security log investigation**.

The biggest takeaway was understanding that SIEM investigation is not simply about searching for individual events. An analyst needs to **extract the right fields, filter relevant activity, pivot across available indicators, and build context around what happened**.

Skills Demonstrated

Splunk Enterprise | SIEM | SPL | Log Ingestion | JSON Log Analysis | Event Searching | Source IP Investigation | User Activity Analysis | Geographic Analysis | SOC Investigation